Exam :- 5

1. Norm-1 and Norm-2 of Vectors.

⇒ Norm :- 1

The L_1 norm of a vector is the sum of absolute values of its components.

$$\|\vec{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

⇒ Norm :- 2

The L_2 norm of a vector is the square root the sum of squares of its components

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$

2. Dot product :-

⇒ The dot product is positive if vectors point in roughly the same direction.

⇒ Zero if perpendicular.

Formula

$$\vec{A} \cdot \vec{B} = \sum_{i=1}^{n} a_i b_i$$

3. Cross Product :-

⇒ The Cross Product of two Vectors A and B in 3D Space gives a new Vector that is

⇒ perpendicular [orthogonal] to both A and B.

⇒ Magnitude proportional to the sine of the angle between them.

$$A \times B = |A||B| \sin(\theta) n$$

4. Matrix addition and Multiplication

⇒ Matrix addition

Ex:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

⇒ Multiplication

Ex:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} [(1\times2)+(2\times1)] & [(1\times0)+(2\times2)] \\ [(3\times2)+(4\times1)] & [(3\times0)+(4\times2)] \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

5. Transpose and Inverse

⇒ **Transpose**

⇒ Flip rows into Columns.

$$(A^T)_{ij} = A_{ji}$$

Ex.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

⇒ **Inverse**

⇒ Defined only for square matrices with non-zero determinat.

Formula

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

$$|A| \neq 0$$

6. Determinat.

⇒ The determinat is a single number calculated from a square matrix.

Formula

$$\det(A) = ad - bc$$

Ex.

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$\det = (2 \times 4) - (3 \times 1)$$
$$= 8 - 3$$
$$= 5$$

7. EigenValues and EigenVectors

i) EigenValues

=> how much the eigenvector is scaled when multipled by the matrix

formula

$$AV = \lambda V$$

Ex.

Image Compression [PCA]

=> how important each pattern is

ii) EigenVectors

=> A Non-Zero Vector that does not change its Change its direction

Ex. Image Compression [PCA]

⇒ main patterns/features in image.